Group 3

Group 3 had a really interesting approach to the project, especially with their use of switches to constrain the LED chasing pattern and automatically reverse it when the next position was blocked. Their design was also separated into individual modules, which made the overall structure of the project clean and organized. One technical question we had was why the slower clock edge needed to be reset whenever the circuit was reset. We also noticed that their counter appeared to start at 1 or 14 rather than using the full 0 to 15 range, which was different from how we approached our design. An alternative approach could have been simplifying some of the counter and clock logic depending on the intended behavior. Compared to our project, their design had more control over the chasing pattern through the switches, which was a really cool twist. Their flowchart also explained the overall design well, although showing large sections of Verilog code during the presentation made some parts harder to follow.

Group 1

We are group 1 and we presented.

Group 5

Group 5 had several interesting additions to their project, especially their use of a multiplexer and the ability to change the speed of the circuit. Their timer acceleration was a cool feature and gave their design more functionality than a basic implementation. One question we had was why they chose the specific clock frequencies they used and how those frequencies affected the behavior of the design. Their counter also operated over a different range rather than simply counting from 0 to 9, which showed another possible approach to implementing the project. Compared to our work, Group 5 focused more on adding different speed options and controlling the timing of the circuit, while our implementation focused more on the flip-flop and shifting behavior. We also thought their slides did a good job of explaining the design without relying as heavily on showing code, which made their presentation easier to follow.